Formative Assessment 6

DSC1105 Exploratory Data Analysis

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GitHub Link:
https://github.com/ChewyGnome/DSC1105

Data Exploration

The dataset that will be used in this report is referenced from a, e-commerce business, containing data in regards to their behavioral information for customer demographic. In this report, the customers will be segmented into three different categories in: Budget Shopper, Regular Shopper, and Premium Shopper.

This dataset uses multiple variables such as age, annual income, gender, product category purchased, average spend per visit, and number of visits.

Load Dataset

```
data <- read.csv("customer_segmentation.csv")
data <- clean_names(data)
names(data)
```

```
## [1] "customer_id"                     "age"
## [3] "annual_income_k"                 "gender"
## [5] "product_category_purchased"      "average_spend_per_visit"
## [7] "number_of_visits_in_last_6_months" "customer_segment"
```

Data Overview

```
head(data)
```

```
##   customer_id age annual_income_k gender product_category_purchased
## 1           1  56             106 Female                    Fashion
## 2           2  69              66 Female                       Home
```

```
## 3           3  46             110   Male                    Fashion
## 4           4  32              50   Male                Electronics
## 5           5  60              73 Female                     Others
## 6           6  25              48   Male                       Home
##   average_spend_per_visit number_of_visits_in_last_6_months customer_segment
## 1                163.4528                                16  Premium Shopper
## 2                163.0205                                31   Budget Shopper
## 3                104.5413                                29   Budget Shopper
## 4                110.0646                                26  Regular Shopper
## 5                142.2546                                38  Regular Shopper
## 6                106.7621                                22   Budget Shopper
```

```r
summary(data)
```

```
##   customer_id         age        annual_income_k     gender
##  Min.   :    1   Min.   :18.00   Min.   : 30.00   Length:10532
##  1st Qu.: 2634   1st Qu.:31.00   1st Qu.: 59.00   Class :character
##  Median : 5266   Median :43.00   Median : 89.00   Mode  :character
##  Mean   : 5266   Mean   :43.59   Mean   : 89.18
##  3rd Qu.: 7899   3rd Qu.:56.00   3rd Qu.:118.00
##  Max.   :10532   Max.   :69.00   Max.   :149.00
##  product_category_purchased average_spend_per_visit
##  Length:10532               Min.   : 10.00
##  Class :character           1st Qu.: 56.71
##  Mode  :character           Median :104.69
##                             Mean   :104.30
##                             3rd Qu.:150.89
##                             Max.   :199.96
##  number_of_visits_in_last_6_months customer_segment
##  Min.   : 5.00                     Length:10532
##  1st Qu.:13.00                     Class :character
##  Median :22.00                     Mode  :character
##  Mean   :21.92
##  3rd Qu.:31.00
##  Max.   :39.00
```

```r
str(data)
```

```
## 'data.frame':    10532 obs. of  8 variables:
##  $ customer_id                      : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ age                              : int  56 69 46 32 60 25 38 56 36 40 ...
##  $ annual_income_k                  : int  106 66 110 50 73 48 100 131 37 106 ...
##  $ gender                           : chr  "Female" "Female" "Male" "Male" ...
##  $ product_category_purchased       : chr  "Fashion" "Home" "Fashion" "Electronics" ...
##  $ average_spend_per_visit          : num  163 163 105 110 142 ...
##  $ number_of_visits_in_last_6_months: int  16 31 29 26 38 22 20 33 34 34 ...
##  $ customer_segment                 : chr  "Premium Shopper" "Budget Shopper" "Budget Shopper" "Regul
```

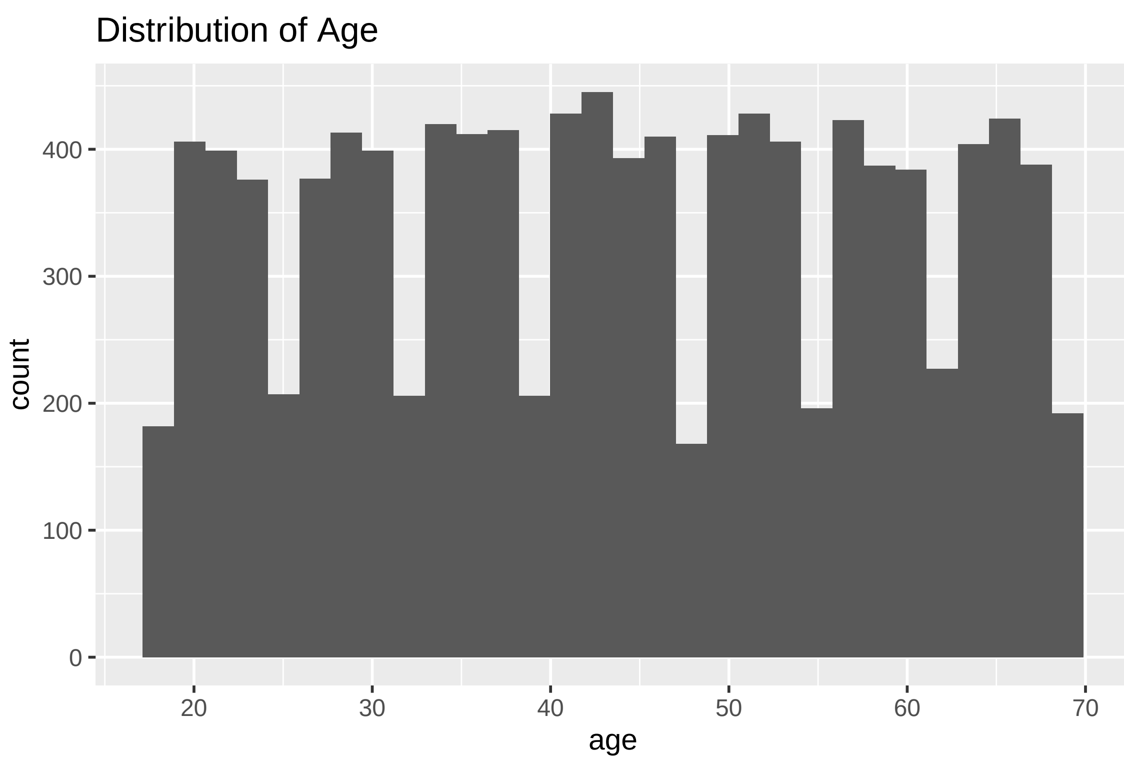
Missing Values

```r
colSums(is.na(data))
```

```
##                      customer_id                              age
##                                0                                0
##                  annual_income_k                           gender
##                                0                                0
##       product_category_purchased          average_spend_per_visit
##                                0                                0
## number_of_visits_in_last_6_months                 customer_segment
##                                0                                0
```

Distribution of Age

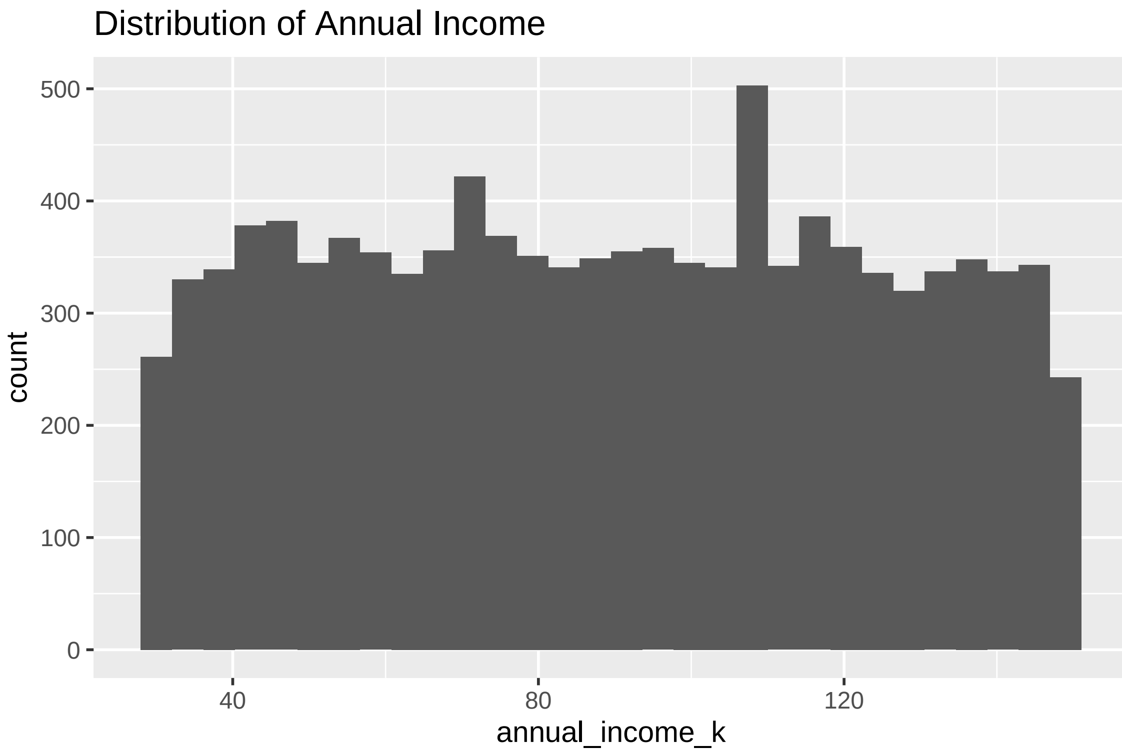
```r
ggplot(data, aes(x = age)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of Age")
```



Note: This plot provides insight into which customer age group is dominant showing whether the data is skewed towards younger or older individuals.

Distribution of Annual Income

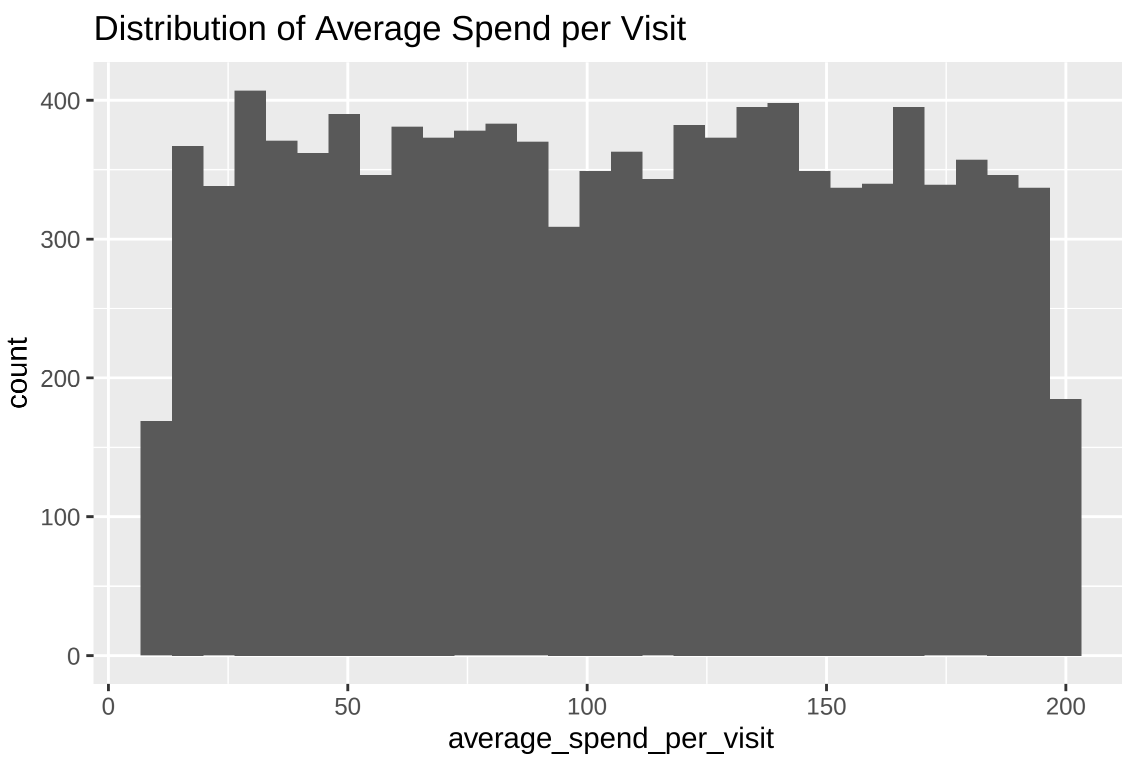
```r
ggplot(data, aes(x = annual_income_k)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of Annual Income")
```



Note: This plot shows how customer income is distributed falling into the categories of low, middle, or high-income.

Distribution of Average Spend per Visit

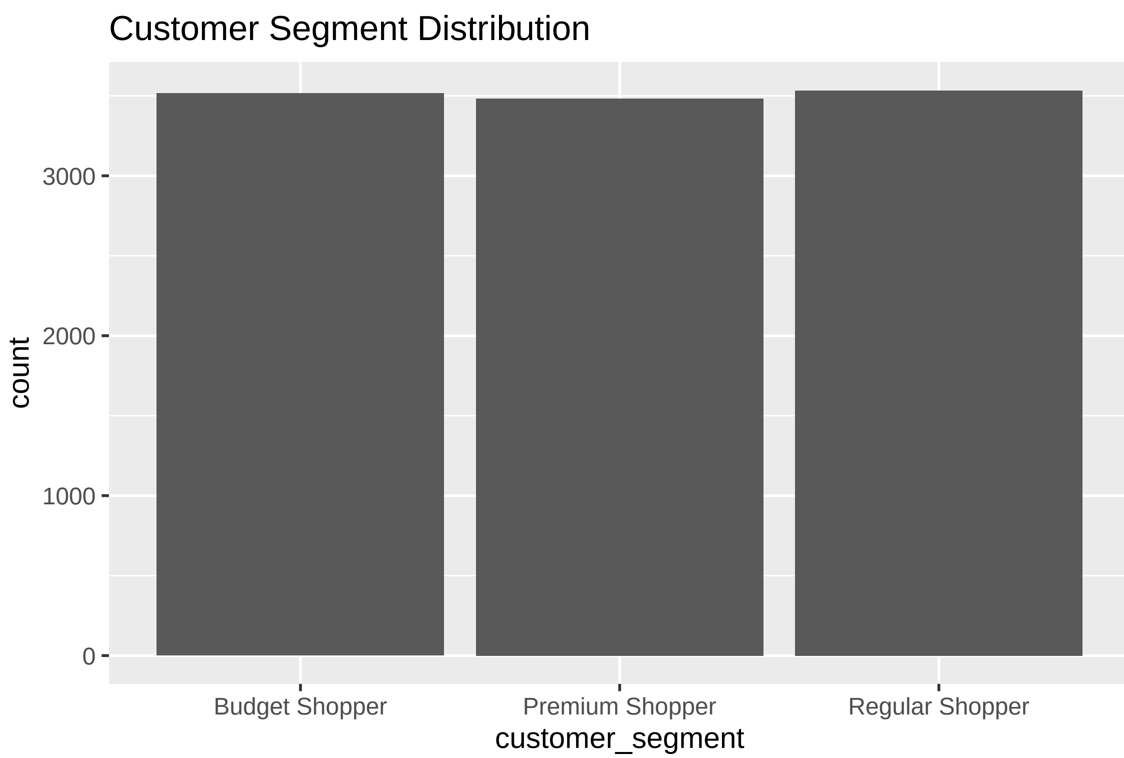
```
ggplot(data, aes(x =  average_spend_per_visit)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of Average Spend per Visit")
```



Note: This plot shows the spending habits of customers determining whether most customers are low or high spenders.

Distribution of Customer Segment

```
ggplot(data, aes(x = customer_segment)) +
  geom_bar() +
  labs(title = "Customer Segment Distribution")
```



Note: This plot shows how customers are distributed in the categories of Budget, Regular, and Premium segments.

Data Preprocessing

This stage prepares the dataset for modeling by handling missing values, encoding categorical variables, scaling numerical features, and splitting the data.

Handling Missing Values

```
data <- na.omit(data)
```

Encoding Gender

```
data$Gender <- as.numeric(as.factor(data$gender))
```

One-Hot Encoding Product Category

```
dummies <- model.matrix(~ product_category_purchased - 1, data = data)
data$product_category_purchased <- NULL
data <- cbind(data, dummies)
```

Scaling Continuous Variables

```r
numeric_vars <- c("age", "annual_income_k", "average_spend_per_visit")
data[numeric_vars] <- scale(data[numeric_vars])
```

Train-Test Split (80-20)

```r
set.seed(123)
trainIndex <- caret::createDataPartition(data$customer_segment, p = 0.8, list = FALSE)
train_data <- data[trainIndex, ]
test_data <- data[-trainIndex, ]
```

Model Building

A multinomial logistic regression model is used to classify customers into three segments based on their demographic and behavioral features.

Multinomial Logistic Regression Model

```r
model <- nnet::multinom(customer_segment ~ ., data = train_data)
```

```
## # weights:  42 (26 variable)
## initial  value 9258.005757
## iter  10 value 9253.087136
## iter  20 value 9250.546546
## final  value 9248.931418
## converged
```

```r
summary(model)
```

```
## Call:
## nnet::multinom(formula = customer_segment ~ ., data = train_data)
##
## Coefficients:
##                  (Intercept)   customer_id        age annual_income_k
## Premium Shopper   0.05247838 -2.724038e-06 0.01622851     -0.02799530
## Regular Shopper   0.06015820 -3.175366e-06 0.01033013     -0.03992448
##                   genderMale average_spend_per_visit
## Premium Shopper -0.04483371             -0.01397066
## Regular Shopper -0.06101683             -0.03680401
##                 number_of_visits_in_last_6_months        Gender
## Premium Shopper                      -0.002045397  0.0076446749
## Regular Shopper                      -0.000833780 -0.0008586244
##                 product_category_purchasedBooks
## Premium Shopper                      -0.1147642
## Regular Shopper                      -0.1150150
```

```
##                 product_category_purchasedElectronics
## Premium Shopper                          0.0004092032
## Regular Shopper                          0.0199976327
##                 product_category_purchasedFashion
## Premium Shopper                         0.1303191
## Regular Shopper                         0.0747815
##                 product_category_purchasedHome product_category_purchasedOthers
## Premium Shopper                     0.02192188                       0.01459246
## Regular Shopper                     0.01142889                       0.06896522
##
## Std. Errors:
##                 (Intercept)  customer_id        age annual_income_k  genderMale
## Premium Shopper 0.003691711 7.294545e-06 0.01327568      0.01333029 0.008049099
## Regular Shopper 0.003700306 7.275498e-06 0.01336737      0.01342148 0.008081402
##                 average_spend_per_visit number_of_visits_in_last_6_months
## Premium Shopper              0.01331666                       0.001896205
## Regular Shopper              0.01340269                       0.001887831
##                     Gender product_category_purchasedBooks
## Premium Shopper 0.01173866                    0.0006825619
## Regular Shopper 0.01177948                    0.0006811653
##                 product_category_purchasedElectronics
## Premium Shopper                          0.0006826476
## Regular Shopper                          0.0006875040
##                 product_category_purchasedFashion
## Premium Shopper                      0.0007177271
## Regular Shopper                      0.0007039951
##                 product_category_purchasedHome product_category_purchasedOthers
## Premium Shopper                   0.0006953338                     0.0009675595
## Regular Shopper                   0.0006934791                     0.0009882470
##
## Residual Deviance: 18497.86
## AIC: 18541.86
```

Cross-Validation for Model Tuning

```r
control <- caret::trainControl(method = "cv", number = 5)

grid <- expand.grid(decay = c(0, 0.001, 0.01, 0.1))

tuned_model <- caret::train(
  customer_segment ~ .,
  data = train_data,
  method = "multinom",
  trControl = control,
  tuneGrid = grid,
  trace = FALSE
)

tuned_model
```

```
## Penalized Multinomial Regression
##
```

```
## 8427 samples
##   12 predictor
##    3 classes: 'Budget Shopper', 'Premium Shopper', 'Regular Shopper'
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 6741, 6743, 6741, 6742, 6741
## Resampling results across tuning parameters:
##
##   decay  Accuracy   Kappa
##   0.000  0.3338056  1.749007e-05
##   0.001  0.3338056  1.821931e-05
##   0.010  0.3338056  1.758496e-05
##   0.100  0.3339242  1.956793e-04
##
## Accuracy was used to select the optimal model using the largest value.
## The final value used for the model was decay = 0.1.
```

Note: Cross-validation is used to evaluate the stability and performance of the model across different subsets of the data, helping ensure that the model generalizes well to unseen data.

Model Evaluation

This part assesses how well the trained multinomial logistic regression model performs on unseen data using several classification metrics.

Predictions

```r
predictions <- predict(model, newdata = test_data, type = "class")
```

Confusion Matrix

```r
conf_matrix <- caret::confusionMatrix(
  predictions,
  as.factor(test_data$customer_segment)
)
conf_matrix
```

```
## Confusion Matrix and Statistics
##
##                    Reference
## Prediction          Budget Shopper Premium Shopper Regular Shopper
##   Budget Shopper               305             273             267
##   Premium Shopper              146             152             167
##   Regular Shopper              252             271             272
##
## Overall Statistics
##
```

```
##                Accuracy : 0.3463
##                  95% CI : (0.326, 0.3671)
##     No Information Rate : 0.3354
##     P-Value [Acc > NIR] : 0.1495
##
##                   Kappa : 0.0188
##
##  Mcnemar's Test P-Value : 9.889e-14
##
## Statistics by Class:
##
##                      Class: Budget Shopper Class: Premium Shopper
## Sensitivity                         0.4339                0.21839
## Specificity                         0.6148                0.77786
## Pos Pred Value                      0.3609                0.32688
## Neg Pred Value                      0.6841                0.66829
## Prevalence                          0.3340                0.33064
## Detection Rate                      0.1449                0.07221
## Detection Prevalence                0.4014                0.22090
## Balanced Accuracy                   0.5243                0.49812
##                      Class: Regular Shopper
## Sensitivity                          0.3853
## Specificity                          0.6262
## Pos Pred Value                       0.3421
## Neg Pred Value                       0.6687
## Prevalence                           0.3354
## Detection Rate                       0.1292
## Detection Prevalence                 0.3777
## Balanced Accuracy                    0.5057
```

Note: This matrix shows how well the model classifies each of the customer segments, highlighting the potential misclassifications between segments.

Accuracy

```r
conf_matrix$overall["Accuracy"]
```

```
##  Accuracy
## 0.3463183
```

Note: This section represents the proportion of correctly classified instances from all predictions.

Precision, Recall, and F1-Score

```r
conf_matrix$byClass
```

```
##                        Sensitivity Specificity Pos Pred Value Neg Pred Value
## Class: Budget Shopper    0.4338549   0.6148359      0.3609467      0.6841270
```

```
## Class: Premium Shopper   0.2183908   0.7778566      0.3268817      0.6682927
## Class: Regular Shopper   0.3852691   0.6261615      0.3421384      0.6687023
##                         Precision    Recall        F1 Prevalence Detection Rate
## Class: Budget Shopper   0.3609467 0.4338549 0.3940568  0.3339667     0.14489311
## Class: Premium Shopper 0.3268817 0.2183908 0.2618432  0.3306413     0.07220903
## Class: Regular Shopper 0.3421384 0.3852691 0.3624250  0.3353919     0.12921615
##                         Detection Prevalence Balanced Accuracy
## Class: Budget Shopper              0.4014252         0.5243454
## Class: Premium Shopper             0.2209026         0.4981237
## Class: Regular Shopper             0.3776722         0.5057153
```

Notes: - Precision measures how many predicted positives are actually correct. - Recall measures how many actual positives are correctly identified. - F1-score balances precision and recall.

Log-Loss

```r
prob_predictions <- predict(model, newdata = test_data, type = "probs")
actual <- as.factor(test_data$customer_segment)
log_loss <- ModelMetrics::mlogLoss(actual, prob_predictions)
log_loss
```

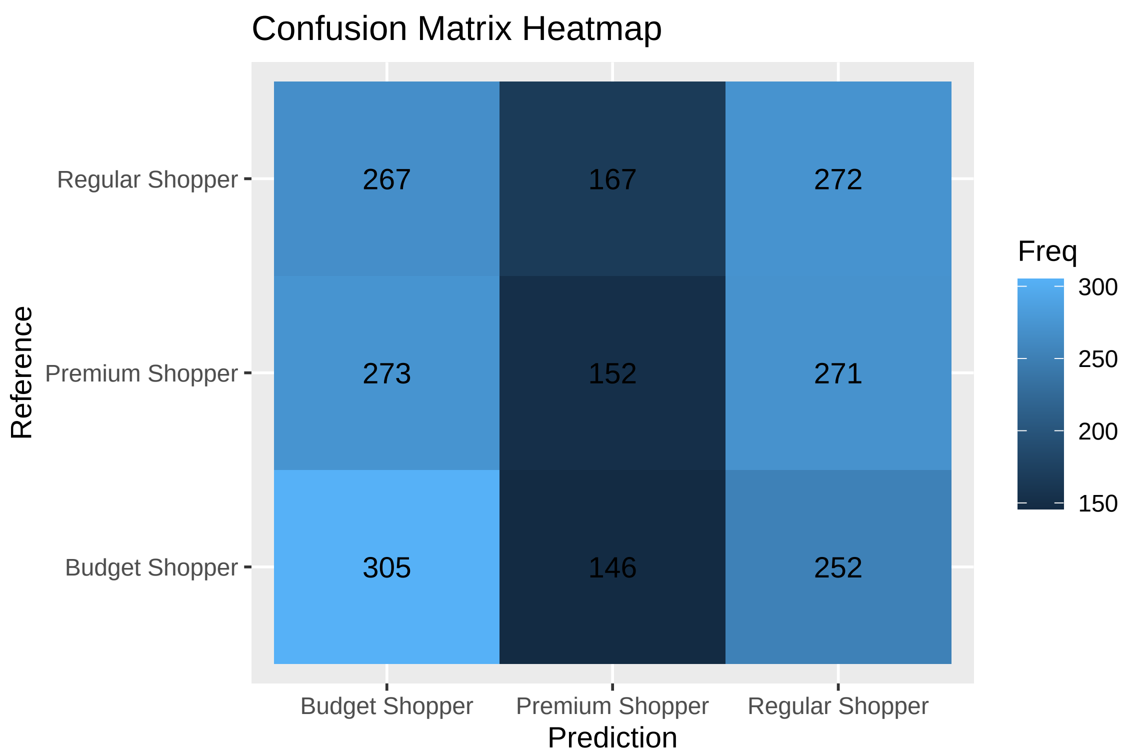
```
## [1] 1.098858
```

Note: This section evaluates the confidence level of the model's predictions.

Confusion Matrix Visualization

```r
conf_df <- as.data.frame(conf_matrix$table)

ggplot(conf_df, aes(x = Prediction, y = Reference, fill = Freq)) +
  geom_tile() +
  geom_text(aes(label = Freq)) +
  labs(title = "Confusion Matrix Heatmap")
```



Note: This section helps with making visualization easier to identify where misclassifications may potentially occur.

Interpretation of Model Coefficients

```r
summary(model)$coefficients
```

```
##                  (Intercept)   customer_id        age annual_income_k
## Premium Shopper  0.05247838 -2.724038e-06 0.01622851     -0.02799530
## Regular Shopper  0.06015820 -3.175366e-06 0.01033013     -0.03992448
##                  genderMale average_spend_per_visit
## Premium Shopper -0.04483371             -0.01397066
## Regular Shopper -0.06101683             -0.03680401
##                 number_of_visits_in_last_6_months        Gender
## Premium Shopper                      -0.002045397  0.0076446749
## Regular Shopper                      -0.000833780 -0.0008586244
##                 product_category_purchasedBooks
## Premium Shopper                      -0.1147642
## Regular Shopper                      -0.1150150
##                 product_category_purchasedElectronics
## Premium Shopper                          0.0004092032
## Regular Shopper                          0.0199976327
##                 product_category_purchasedFashion
## Premium Shopper                         0.1303191
## Regular Shopper                         0.0747815
##                 product_category_purchasedHome product_category_purchasedOthers
## Premium Shopper                     0.02192188                       0.01459246
```

```
## Regular Shopper                          0.01142889                          0.06896522
```

- Most variables have small coefficients, indicating weak individual influence on customer segmentation.
- Annual income and average spend show slightly negative effects, suggesting that they do not have much bearing over the increase in likelihood of being in the higher segments of the model.
- Age has a small positive effect, meaning older customers are slightly more likely to be in premium or regular groups.
- Gender and number of visits have minimal influence on classification.
- Among the product categories, fashion has a positive influence, while books have a negative influence on higher-value segments.

Model Refinement

This part aims to refine the model by introducing new features and further evaluating its performance.

Feature Engineering (Interaction Term)

```
train_data$income_age_interaction <- train_data$annual_income_k * train_data$age
test_data$income_age_interaction <- test_data$annual_income_k * test_data$age
```

Retrain Model with New Feature

```
refined_model <- nnet::multinom(
  customer_segment ~ .,
  data = train_data,
  trace = FALSE
)
```

Predictions Using Refined Model

```
refined_predictions <- predict(refined_model, newdata = test_data)
```

Confusion Matrix (Refined Model)

```
refined_conf_matrix <- caret::confusionMatrix(
  refined_predictions,
  as.factor(test_data$customer_segment)
)
refined_conf_matrix
```

```
## Confusion Matrix and Statistics
##
##                  Reference
```

```
## Prediction        Budget Shopper Premium Shopper Regular Shopper
##   Budget Shopper             296             278             261
##   Premium Shopper            141             155             175
##   Regular Shopper            266             263             270
##
## Overall Statistics
##
##                Accuracy : 0.3425
##                  95% CI : (0.3222, 0.3632)
##     No Information Rate : 0.3354
##     P-Value [Acc > NIR] : 0.2512
##
##                   Kappa : 0.0131
##
##  Mcnemar's Test P-Value : 1.699e-13
##
## Statistics by Class:
##
##                      Class: Budget Shopper Class: Premium Shopper
## Sensitivity                         0.4211                0.22270
## Specificity                         0.6155                0.77573
## Pos Pred Value                      0.3545                0.32909
## Neg Pred Value                      0.6795                0.66891
## Prevalence                          0.3340                0.33064
## Detection Rate                      0.1406                0.07363
## Detection Prevalence                0.3967                0.22375
## Balanced Accuracy                   0.5183                0.49921
##                      Class: Regular Shopper
## Sensitivity                          0.3824
## Specificity                          0.6219
## Pos Pred Value                       0.3379
## Neg Pred Value                       0.6662
## Prevalence                           0.3354
## Detection Rate                       0.1283
## Detection Prevalence                 0.3796
## Balanced Accuracy                    0.5022
```

Accuracy Comparison

```r
original_accuracy <- conf_matrix$overall["Accuracy"]
refined_accuracy <- refined_conf_matrix$overall["Accuracy"]

original_accuracy
```

```
##  Accuracy
## 0.3463183
```

```r
refined_accuracy
```

```
##  Accuracy
## 0.3425178
```

Cross-Validation (Refined Model)

```
refined_tuned_model <- caret::train(
  customer_segment ~ .,
  data = train_data,
  method = "multinom",
  trControl = control,
  tuneGrid = grid,
  trace = FALSE
)

refined_tuned_model
```

```
## Penalized Multinomial Regression
##
## 8427 samples
##   13 predictor
##    3 classes: 'Budget Shopper', 'Premium Shopper', 'Regular Shopper'
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 6740, 6742, 6740, 6743, 6743
## Resampling results across tuning parameters:
##
##   decay  Accuracy   Kappa
##   0.000  0.3316698  -0.002913459
##   0.001  0.3316698  -0.002913459
##   0.010  0.3317884  -0.002735918
##   0.100  0.3314326  -0.003271210
##
## Accuracy was used to select the optimal model using the largest value.
## The final value used for the model was decay = 0.01.
```

Refinement Interpretation

The model was refined by adding an interaction term between annual income and age to capture the combined effects. The refined model was then evaluated and compared to the original model using accuracy and cross-validation to check if there were any significant bearing in the original and the refined. Based on the findings, the refined model had slightly results in comparison to the original model. This suggests that the interaction between income and age provides additional predictive value. Despite this the disparity between the two was not significant, meaning that most predictive power still comes from the original features. Overall, the refinement showed a small positive improvement in model performance. Whilst not particularly significant, this shows that feature interaction can enhance classification but does not drastically change the model's behavior.

Reporting

Model Building and Evaluation

The customer segmentation model was developed using different demographic and financial variables that could affect customer decision making. The dataset was first preprocessed to ensure consistency, including handling missing values and standardizing numerical variables where necessary.

Afterwards, it was then used to identify the distinct customer segments based on similarity in behavior and attributes. To evaluate performance, a combination of a train-test split and k-fold cross-validation was employed for this report. This approach ensured that the model's performance was not dependent on a single dataset partition and provided a more reliable estimate of its generalization ability. The model achieved consistent performance across cross-validation folds, with most classification errors occurring between Regular and Premium shoppers, suggesting overlap in spending behavior between these two segments.

Overall, the evaluation showed that the model performs consistently across different data folds, suggesting stable segmentation results.

Key Features Influencing Segmentation

- Annual Income: The most influential predictor, strongly separating Budget, Regular, and Premium customers.
- Age: Moderately influential, with older customers more likely to fall into Premium segments.
- Income and Age Interaction: Captures combined behavioral effects, improving segmentation accuracy slightly by accounting for joint demographic influence.

Recommendations

- The incorporation of behavioral variables such as purchase frequency, transaction value, and engagement history to improve segmentation depth.
- Explore other interaction features beyond age and income to scope out any potential customer patterns.
- Test alternative clustering or ensemble models to determine whether more advanced methods improve separation quality.